

TEST OF MATHEMATICS FOR UNIVERSITY ADMISSION

PAPER 2: MATHEMATICAL REASONING

Practice paper · Set 15

75 minutes

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but do not open the question paper until you are told that you may do so.

A separate answer sheet is provided for this paper. You require a soft pencil and an eraser. Please complete the answer sheet with your candidate number, centre number, date of birth and full name.

This paper contains **20 questions**. For each question, choose the **one** answer you consider correct and record your choice on the separate answer sheet. If you make a mistake, erase thoroughly and try again.

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all 20 questions. Each question is worth one mark.

You can use the question paper for rough working or notes, but no extra paper is allowed.

You must complete the answer sheet within the time limit.

Calculators and dictionaries are NOT permitted.

There is no formulae booklet for this test.

Please wait to be told you may begin before turning this page.

This is a practice paper generated from a question bank written to the TMUA content specification. It is not an official UAT-UK paper. Answers and worked solutions are in the accompanying solutions booklet.

BLANK PAGE

1 Solve $\log_2 x + \log_4 x = 6$. (You may use $\log_4 x = \frac{1}{2}\log_2 x$.)

A $x = 8$

B $x = 2^6$

C $x = 16$

D $x = 64$

E $x = 4$

F $x = 32$

2 How many solutions does $\sin 2x = \frac{1}{2}$ have for $0 \leq x < 2\pi$?

A 2

B 6

C 4

D 8

E 1

F 3

- 3** Given that $\int_0^4 f(x) \, dx = 10$, find $\int_0^4 (f(x) + 3x) \, dx - \int_4^0 f(x) \, dx$.
- A** 20
- B** 0
- C** 14
- D** 44
- E** 24
- F** 34

- 4 Which of the following is a counterexample to the claim **if n is a positive integer, then $2^n + 1$ is prime**?
- A $n = 8$
 - B $n = 1$
 - C $n = 4$
 - D $n = 3$
 - E $n = 2$

- 5 Which of the following statements is/are true?
- I** Every rhombus is a parallelogram.
 - II** Every rectangle is a rhombus.
 - III** The diagonals of a kite are perpendicular.
- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

- 6** It is given that for all real x : if $f(x) > 0$ then $g(x) > 0$. It is also known that $g(2) = -1$. What can be deduced?
- A** $f(2) < 0$
 - B** $f(2) = 0$
 - C** $f(2) > 0$
 - D** $f(2) \leq 0$
 - E** nothing about $f(2)$

- 7 A geometric series has first term $a \neq 0$ and common ratio r . P : ' $-1 < r < 1$ '. Q : 'the series has a sum to infinity'. Which one of the following is true?
- A P is necessary and sufficient for Q.
 - B P is necessary but not sufficient for Q.
 - C P is sufficient but not necessary for Q.
 - D P is neither necessary nor sufficient for Q.

8 The sum of 5 consecutive odd numbers is S . Which of the following is always a factor of S ?

A 4

B 3

C 10

D 25

E 15

F 5

- 9** Let P : 'the quadrilateral $ABCD$ is a square' and Q : 'the quadrilateral $ABCD$ has four equal sides'. Which one of the following is true?
- A** P is necessary and sufficient for Q .
 - B** P is necessary but not sufficient for Q .
 - C** P is sufficient but not necessary for Q .
 - D** P is neither necessary nor sufficient for Q .

10 A student solves $x^2 - 5x + 6 > 0$ as follows.

I $x^2 - 5x + 6 = (x - 2)(x - 3)$.

II So $(x - 2)(x - 3) > 0$.

III So $x - 2 > 0$ and $x - 3 > 0$.

IV So $x > 3$.

Which of the following is correct?

- A** The first error is on line III: a product is positive when both factors are positive **or** both are negative, so $x < 2$ is missed.
- B** The solution is completely correct.
- C** The first error is on line I.
- D** The first error is on line IV: it should be $x > 2$.
- E** The first error is on line II.

- 11** Two distinct lines have equations $y = m_1x + c_1$ and $y = m_2x + c_2$. Which of the following statements is/are true?

- I** If $m_1m_2 = -1$ then the lines meet at exactly one point.
- II** If $m_1 = m_2$ then the lines do not meet.
- III** If the lines meet on the y-axis then $c_1 = c_2$.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

12 Consider the statement: ****if a and b are irrational, then $a + b$ is irrational.**** Which of the following pairs is a counterexample?

A $a = 1, b = -1$

B $a = \sqrt{2}, b = -\sqrt{2}$

C $a = \pi, b = \sqrt{2}$

D $a = \sqrt{2}, b = 2\sqrt{2}$

E $a = \sqrt{2}, b = \sqrt{3}$

13 Which of the following statements is/are true?

I For every real number x , there exists a real number y such that $y > x$.

II There exists a real number y such that for every real number x , $y > x$.

III For every real number x , there exists a real number y such that $xy = 1$.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

14 Let $f(x) = \sqrt{x-3}$ for $x \geq 3$ and $g(x) = x^2 + 3$. Which of the following statements is/are true?

I $g(f(x)) = x$ for all $x \geq 3$

II $f(g(x)) = x$ for all real x

III $f(g(x)) = |x|$ for all real x

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

15 a and b are positive real numbers. Which of the following statements is/are true?

I If $a^2 = b^2$ then $a = b$.

II If $2^a = 2^b$ then $a = b$.

III If $a^b = 1$ then $a = 1$.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

- 16** The expansion of $\left(2x - \frac{1}{x^2}\right)^9$ contains a term in x^k for each of which of the following values of k ?
- I** $k = 0$
 - II** $k = 3$
 - III** $k = -6$
- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

- 17 Claim: for all positive real x , $x + \frac{1}{x} \geq 2$.
- I Suppose $x + \frac{1}{x} \geq 2$.
 - II Multiply by $x > 0$: $x^2 + 1 \geq 2x$.
 - III So $x^2 - 2x + 1 \geq 0$, i.e. $(x - 1)^2 \geq 0$, which is true.
 - IV Hence the claim is proved.

Which of the following best describes the argument?

- A The argument assumes what it is trying to prove (line I); it would be valid if the steps were reversed, starting from $(x - 1)^2 \geq 0$.
- B The argument is completely correct as written.
- C The first error is on line II: multiplying by x reverses the inequality.
- D The first error is on line III.
- E The claim is false.

18 $f(x) = x^2 + 1$. Which of the following statements is/are true?

I $f(f(x)) = x^4 + 2x^2 + 2$ for all x

II $f(x + 1) = f(x) + 1$ for all x

III $f(2x) = 4f(x) - 3$ for all x

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

19 Three statements are made about a positive integer N :

- N is a perfect square.
- N is a multiple of 6.
- N is less than 50.

Exactly two of the statements are true. Which of the following could NOT be N ?

A 144

B 12

C 36

D 48

E 16

F 25

- 20** How many ordered pairs of positive integers (m, n) satisfy $\frac{1}{m} + \frac{1}{n} = \frac{1}{4}$?
- A** 4
 - B** 3
 - C** 2
 - D** 6
 - E** 8
 - F** 5

END OF TEST